LETTER

doi:10.1038/nature22321

Topological defects control collective dynamics in neural progenitor cell cultures

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Cultured stem cells have become a standard platform not only for regenerative medicine and developmental biology but also for biophysical studies. Yet, the characterization of cultured stem cells at the level of morphology and of the macroscopic patterns resulting from cell-to-cell interactions remains largely qualitative. Here we report on the collective dynamics of cultured murine neural progenitor cells (NPCs), which are multipotent stem cells that give rise to cells in the central nervous system[1]. At low densities, NPCs moved randomly in an amoeba-like fashion. However, NPCs at high density elongated and aligned their shapes with one another, gliding at relatively high velocities. Although the direction of motion of individual cells reversed stochastically along the axes of alignment, the cells were capable of forming an aligned pattern up to length scales similar to that of the migratory stream observed in the adult brain[2]. The two-dimensional order of alignment within the culture showed a liquid-crystalline pattern containing interspersed topological defects with winding numbers of +1/2 and −1/2 (half-integer due to the nematic feature that arises from the head–tail symmetry of cell-to-cell interaction). We identified rapid cell accumulation at +1/2 defects and the formation of three-dimensional mounds. Imaging at the single-cell level around the defects allowed us to quantify the velocity field and the evolving cell density; cells not only concentrate at +1/2 defects, but also escape from −1/2 defects. We propose a generic mechanism for the instability in cell density around the defects that arises from the interplay between the anisotropic friction and the active force field.

We used NPC culture established from fetal mice, which can survive and proliferate (that is, undergo cell division) for numerous cycles on a dish coated with laminin (which allows the cells to attach to the dish)[6]. At low density, cells tended to move in an amoeba-like fashion, so with no apparent order but with occasional attachment between cells (Fig. 1a, Supplementary Video 1). In the high-density regime, on the other hand, cells aligned with each other according to their elongated shape (Fig. 1a), which led to a macroscopic pattern (Fig. 1b, Supplementary Video 2). The alignment pattern evolved in time following the growth of the cell density, which was captured by quantifying the nematic order parameter at a fixed system size (Extended Data Fig. 1a and b). We found points in the pattern at high density where the cellular alignment was disrupted; these points are called topological defects. The defects observed in this system are characterized by the winding numbers $\pm1/2$; the alignment field around the defects suffer plus or minus half of a full rotation on an enclosing counterclockwise path (Fig. 1c). Defects with winding numbers $\pm1/2$ are precisely the signature of underlying nematic order. If the system is polar, asters and vortices (with winding number +1) will appear instead[7].

To obtain the trajectory of single cells and quantify the cell density (Fig. 2a), we established a cell line that stably expresses a fluorescent nucleus marker, H2B-mCherry. The single-cell tracking showed that local alignments at the high-density regime involved rapid motion (about 40 μm h^{-1}, Extended Data Fig. 2a and b), mostly in the locally aligned direction, whereas in the lower-density stage the motion looked more random (Fig. 2b). From the live images it is clear that NPCs are able to slide and pass by each other even in the seemingly dense-packed configuration (Supplementary Video 3), indicating that the interaction between the directions of cell velocity is close to ideally nematic (Extended Data Fig. 2c). The cell interactions and migrations were dependent on the concentration of laminin; large cell aggregates formed when the concentration was too low (Extended Data Fig. 3a). Consistent with the fact that the bipolar cell shape is supported by microtubules, treatment with nocodazole (which blocks microtubule polymerization) was also capable of disrupting the nematic pattern (Extended Data Fig. 3b).

We quantified the timescale involved in the cell motion by calculating the autocorrelation functions from the single-cell trajectories (see Supplementary Information). The autocorrelation of the direction of motion at high density fitted well with an exponential function, which is a feature of a time-independent stochastic process (Fig. 2c, time frames C and D). To gain further insight into the properties of the motion, we calculated the nematic autocorrelation function of the direction of motion. In contrast to the autocorrelation, the nematic autocorrelation at high density approached zero very slowly with respect to the time interval (Fig. 2d, time frames C and D), but less so at lower density (time frames A and B). This indicates that the cells are changing their direction of motion 180° rather than randomly walking in two dimensions, as also seen in the distribution of angle relative to the longitudinal direction of motion of each cell (Extended Data Fig. 2b). Taken together, the timescale extracted from the autocorrelation can be compared with a one-dimensional velocity reversal process, especially at the high-density culture, where we find the average switching time to be about 3 h (inset to Fig. 2c).

In adult rodent brains, neuroblast cells—a premature form of neurons—continuously migrate out of the subventricular zone in a millimetre-order collective cellular line, which is a phenomenon known as the rostral migratory stream (RMS)[2]. Similar to our observations for NPCs, the motion of the cells in the RMS has been shown to be bipolar within the average transport towards the olfactory bulb[8]. Although our NPCs are premature compared to neuroblasts (that is, they are negative in neuroblast marker PSA-NCAM; Extended Data Fig. 3c), we found that the knockdown of NCAM (one of the cell-adhesion proteins that is critical in RMS), disrupted the alignment patterns, resulting in disordered dynamics (Extended Data Fig. 3d, Supplementary Video 4). We further asked whether a thin lane set by a boundary condition[9,10] could induce NPCs to align on a length scale that is as long as the RMS. Indeed, under a lane boundary condition set by scraping the surface of the culture substrate, the cells persistently aligned over a distance greater than 2 mm while undergoing bidirectional motion, thus mimicking the thin and long pattern seen in the real brain[2] (Fig. 3, Supplementary Video 5).

A neural progenitor in a developing embryo, on the other hand, is known to move its nucleus up and down within its thin cell body

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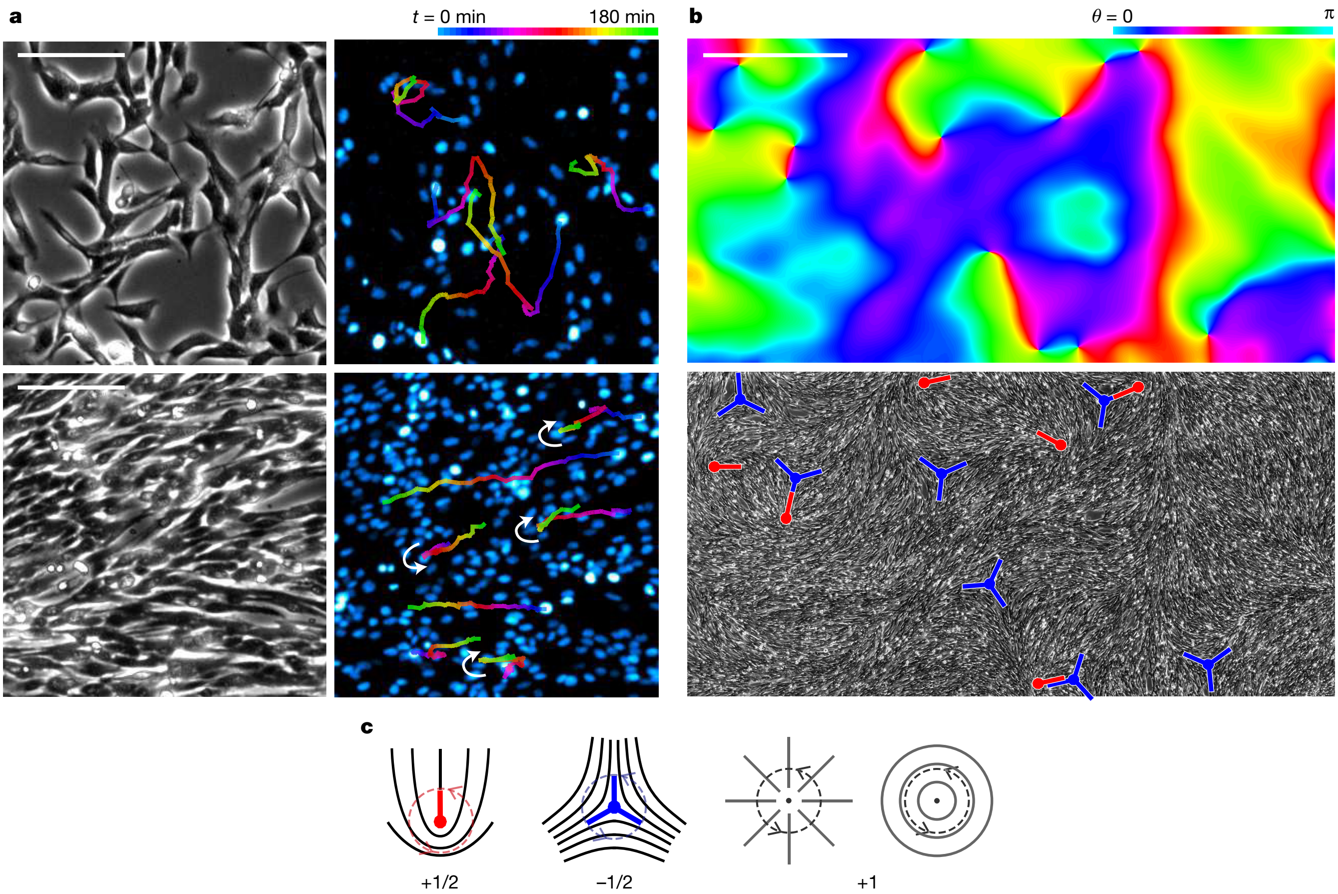

Figure 1 | Two-dimensional patterning and collective motion of NPCs. **a**, Imaging at the single-cell level of the two-dimensional culture. The top panels show low-density (1,100 cells mm^{-2}) cells observed in phase contrast (left) and fluorescent channel (right, nucleus marker in pseudo-colour). Lines in the right panels are the typical trajectories of single cells. The bottom panels show high-density (3,000 cells mm^{-2}) cells. The white arrows (right bottom panel) indicate velocity reversal of the tracked cells. Scale bar, 100 μm. **b**, Large-field image of NPC culture. Colour indicates the angle of the local alignment calculated using the tensor method (see Supplementary Information) from the phase contrast image (bottom). Singular points where all colours meet in the top figure are the topological defects, as shown explicitly in the bottom image with the winding numbers +1/2 (red) and −1/2 (blue) indicated. Scale bar, 1 mm. **c**, Types of topological defects characterized by the winding numbers.

according to the cell cycle phase, which is known as the interkinetic nuclear migration[11]. To see whether the direction switching is related to cell cycle, we used the cell-cycle marker Fucci[12], and found that the switching of direction occurs within the same G1 or G2 cell cycle phase (Supplementary Video 6). This is in contrast to the interkinetic motion seen *in vivo*, where the motion of the nucleus is directed to the basal end (G1) or the apical end (G2) of the NPC, depending on its cell cycle phase[11], although there was a difference in the timescale of switching between the two phases (Extended Data Fig. 4a–c)[13].

Active nematic systems are predicted to have spatio-temporal correlations between densities and strengths of nematic order that are distinct from those of equilibrium systems[3–5]. By picking out areas that do not contain defects[14], we confirmed giant number fluctuation and quasi-long-range order (Extended Data Fig. 5a and b) in the high-density NPC culture. In the same regions, we calculated the spatial correlations of cell density and alignment fluctuations (Extended Data Figs 6 and 7a–f) and found that the fluctuations of density and alignment were correlated (see Supplementary Information). The correlations were positive in the first and third quadrant when seen as a function of two-dimensional displacements, indicating that the system is an extensile active nematic system[3,5,15]. This feature of the correlation was flipped and weaker in the case of a cultured myoblast cell line C2C12, which also shows a nematic pattern (Extended Data Fig. 8a and b). We further found that the spatial correlation of the cell density was anisotropic when the typical length scale of correlation in the direction parallel to the alignment was larger than that in the direction perpendicular to it (Extended Data Fig. 6).

A clear signature of 'activeness' in a macroscopic system can be found at the topological defects. It has been observed both in experiment and simulation[16–25] that +1/2 defects typically undergo spontaneous motion in active nematic systems, which is in contrast to defects in equilibrium liquid crystals[26]. Topological defects in the biological context have been observed in whorled grain in wood[17] as well as in tube formation in embryonic development. Examples of defects yielding non-equilibrium collective dynamics include *in vitro*[21,22,27] and *in vivo*[28] cytoskeleton dynamics, bacterium flows in liquid crystal mixtures[23,25], colony growth dynamics[24], and the cAMP patterning of aggregating social amoeba[29].

As has been reported in other works on cell cultures[9,30,31], we clearly observed +1/2 and −1/2 defects in the dense NPC culture (Fig. 4a). We found slow motions of defects (3.7 μm h^{-1} on average) including the merging of defects with opposite winding numbers (Fig. 4b). In Fig. 4c, we show the local nematic order around the topological defects calculated from the displacement of cells inside small regions. Figure 4d shows the net velocity field calculated from the same data, indicating that there is a flow of cells pointing in the direction of the front of the comet shape in +1/2 defects (Fig. 4c), whereas no clear directionality exists in the −1/2 defects. The direction of the defect motion in the NPCs was opposite to that for passive liquid crystals, C2C12 (Fig. 4e), and the confluent fibroblasts reported in ref. 31.

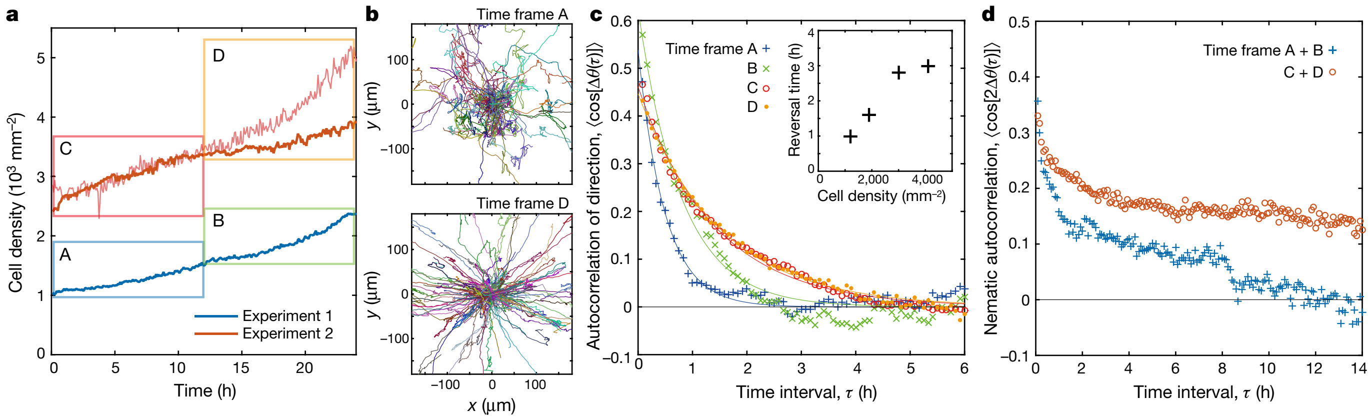

Figure 2 | Single-cell tracking reveals the transition from random to bipolar motion at high density. a, Time dependence of cell density obtained from two 24-h time-lapse videos by counting cell nuclei in the whole images. The thin red line corresponds to the cell density estimated by another method for experiment 2 (integrating the fluorescent signal intensity of H2B-mCherry). At high density (time frame D), the method of nuclei counting starts to fail owing to cell accumulation. **b**, The *x*–*y* coordinates of the trajectories of single cells, shown in different colours, within the time frames A and D from **a**. The starting points of each track were set at the origin in the plots. **c**, Autocorrelation of the instantaneous direction of motion (θ) generated from 61 (for time frames A and B) and 169 (for time frames C and D) randomly tracked cells as a function of time interval τ. Angle brackets indicate ensemble averages, and $\Delta\theta(\tau)$ is the change in the direction of motion of the same cell through time interval τ. The inset shows the velocity reversal time estimated by fitting the autocorrelation data with an exponential function (solid lines; see Supplementary Information). **d**, Nematic autocorrelation of the instantaneous direction of motion generated from the same data as **c** but with pairs of time frames combined to quantify behaviour over a long period.

From the net velocity field around the $+1/2$ defect, we found that the average velocity of cells in the direction of the defect motion was faster at the back of the comet than at the front (Fig. 4f). This indicates that the cells flowing in from the back of the comet are blocked at the front, which should result in cell accumulation around the defect core. Indeed, we observed that cells around the $+1/2$ topological defects tended to accumulate rapidly, forming a three-dimensional mound of cells growing out of the observed plane (Fig. 4g, Supplementary Video 7). The time dependence of cell density includes the effect of cell division, which is happening globally (Extended Data Fig. 9a). To eliminate the effect of uniform cell growth, we normalized the sum of the nucleus reporter intensity using its global spatial average to obtain ρ. An increasing (decreasing) ρ under this normalization will indicate that the cells are accumulating (depleting) over time in that region compared with the global average.

The radius-dependent average cell density, $\rho(R, t)$, was calculated for different diameters R surrounding the defects using the normalized density (see Supplementary Information for detail). Strikingly, we found not only that accumulation is occurring at the $+1/2$ defects, but also that the density around the $-1/2$ defects are decreasing relative to the average growth (Fig. 4a, Supplementary Video 8). The relative growth and escape rates of density near the defect centre (Fig. 4h and i and Extended Data Fig. 9b) were comparable or larger than the global cell growth rate (0.03–0.035 h^{-1}, Fig. 2a), which suggests that local fluctuations in cell growth will be too small to account for the density evolution, and that there should be a substantial cell flow into and out from the $+1/2$ and $-1/2$ defects, respectively. We also note that there is no sign of local differentiation near the defects (Extended Data Fig. 9c), ruling out the possibility of a drastic change in cell properties inducing flow. We conducted the same analysis for the defects in the C2C12 cell culture and found no evidence of relative density growth or decrease (Extended Data Fig. 10).

To understand the cell velocity profile and the flow-blocking effect amounting to cell accumulation around the defects, we consider the theoretical setup using the Q-tensor representation for nematic order. The linearized time-evolution equation of the over-damped net velocity field[3] can be written as

$$\gamma v = -\zeta \nabla \cdot Q \tag{1}$$

Here, the net velocity field of the cells $\boldsymbol{v}$ and the 2×2 nematic order tensor Q are defined locally as functions of space. The right-hand side of the equation is the active force[32], which is the lowest-order component of force that cannot derive from equilibrium free energy. The activity

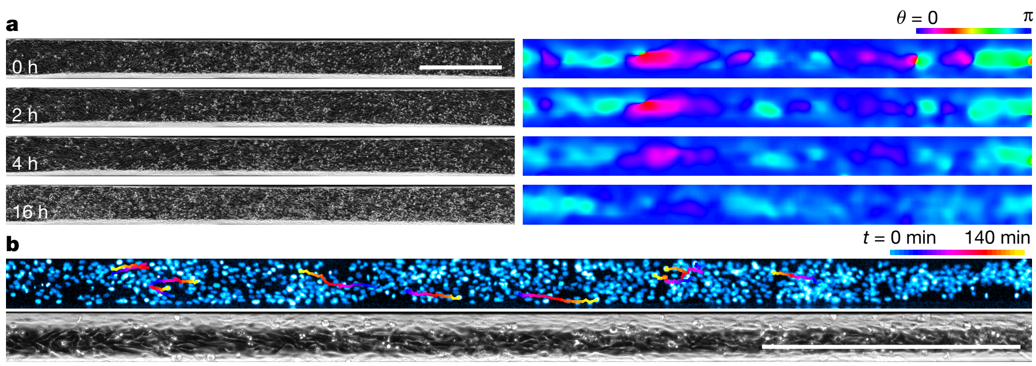

Figure 3 | Long-range nematic order induced by artificial boundaries. a, Culturing NPCs on a plastic substrate with two parallel scrapes (240-μm interval) causes long-range (>3 mm) ordering that is retained for a long time once it is formed. The left panels show phase contrast images. The right panels show local alignments obtained using the tensor method. **b**, Cells migrating bidirectionally within the thin (95 μm) thread whose path persists for longer than 2 mm. The top panel shows a fluorescent image of a cell nucleus with typical trajectories. The bottom panel shows its phase contrast image. See Supplementary Video 5. Scale bars, 500 μm.

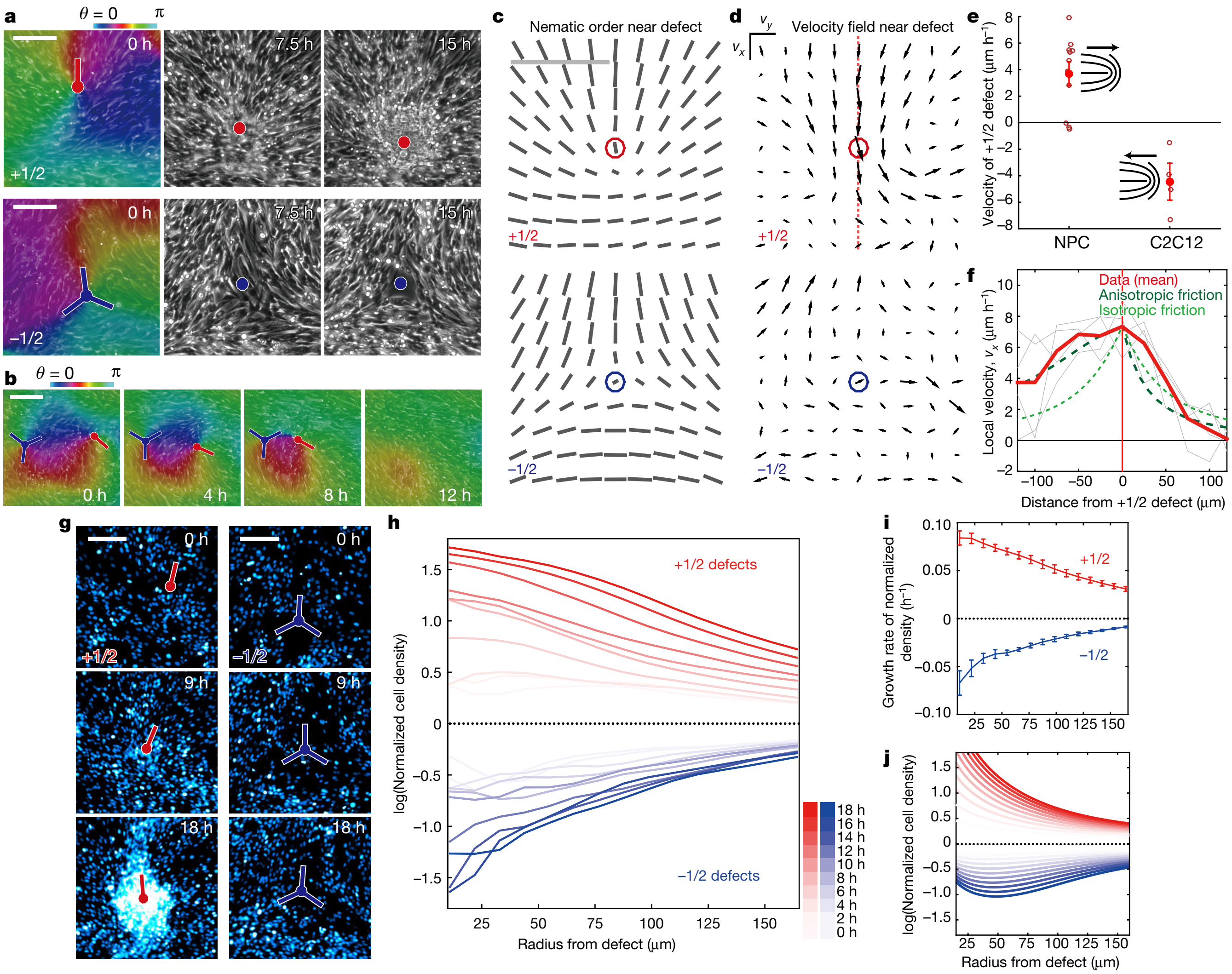

Figure 4 | Active topological defects. a, Typical time course of topological defects observed at a fixed position on glass at high cell density. The red and blue spots correspond to the +1/2 and −1/2 topological defects, respectively. **b**, Merging of +1/2 and −1/2 defects resulting in annihilation. **c**, Local nematic order (length in arbitrary units) and alignment direction obtained by particle tracking. Circles represent the position of the defects. Average of four (+1/2) and five (−1/2) independent defects over the time frames (about 16 h each) before cell accumulation becomes noticeable. **d**, Local velocity field obtained from the same data. The lattice points correspond to those in **c**. The lengths of the arrows are proportional to the velocity, with the same units in the top and bottom figures. **e**, Velocity of +1/2 defects in NPC and myoblast culture (C2C12). Error bars are the standard error of the mean. **f**, Local velocity in the downward direction (v_x) along the centre line (red dots) in **d**. Grey lines show the velocity profiles of individual defects. **g**, Time course of +1/2 (left) and −1/2 (right) topological defects in the fluorescent channel (pseudo-colour). **h**, The logarithm of the cell density ρ_+ (for the +1/2 defect) and ρ_- (for the −1/2 defect) inside a circle centred around topological defects as a function of the radius of the circle R and time t is shown: $\log[\rho_\pm(R, t)]$. Cell density was calculated from the fluorescent signal of the nuclei marker, and normalized by the time-dependent (growing) cell density after subtracting background (see Supplementary Information). Cell density was averaged over four and five defects for +1/2 and −1/2 defects, respectively, within time points where the three-dimensional structure has not become noticeable. **i**, Growth rate of normalized cell density as a function of the radius. Error bars are fitting errors. **j**, Growth of normalized cell density obtained from theory using parameters estimated in **f** and the nematic order around defects. All scale bars, 100 μm.

parameter ζ defines how far from equilibrium the system is[15], and is taken to be positive (extensile) for consistency with the correlation in the density–alignment fluctuations (Extended Data Fig. 6) and the direction of the velocity field around the defects (Fig. 4d). The friction γ that is experienced by the collective cell flow originates from the stochastic switching of the direction of motion, cell-to-cell interaction, and the adhesive interaction between the laminin-coated substrate and the cells. The key assumption is that γ is also a locally defined tensor that depends on the alignment; it includes a term proportional to Q in the leading order as follows: $\gamma = \gamma_0 I - \Delta Q$, where the first term describes the isotropic friction, and the second term with $\Delta > 0$ describes the case of friction being larger against a force that is perpendicular to cell alignment.

We assume specific forms of the Q-tensors corresponding to the alignment field around ±1/2 topological defects, with local nematic order depending only on the distance from the core of the defect. By solving equation (1) and using the length scale in the defect obtained from experimental data (Extended Data Fig. 9b), we calculated the velocity field around the +1/2 defect on the centre line. We see in Fig. 4f that the isotropic case ($\Delta = 0$) already gives a finite velocity, which indicates that the active force alone can drive the motion of defects. However, the velocity profile in the isotropic case is symmetric in the front and the back with respect to the defect, meaning that the inflow and outflow are balanced at the defect core. In contrast, the velocity profile in the anisotropic case ($\Delta = 0.7\gamma_0$) correctly captures the feature observed in the experimental data (Fig. 4f), which suggests that the blocking by the strong anisotropic friction can indeed cause the inflow towards the defect to be larger than the outflow. Using the parameters from Fig. 4f, we numerically obtained the time-evolution of the density that follows equation (1) for both $\pm 1/2$ defects (Fig. 4j) and found qualitative agreement with the observations, indicating that the active force and anisotropic friction also explain the cell density decrease at $-1/2$ defects.

A challenge in biophysics is to understand the laws behind the seemingly extraordinary collective dynamics observed *in vivo*. Since collective cell phenomena are typically out of equilibrium, the dynamics is not necessarily explicable by clear principles such as free energy minimization. Here we observed and analysed the extensile features of the active nematics of NPC culture, and showed that they are capable of aligning for a large length scale under confinement in thin stripes, resembling the behaviour of cells in the migratory streams found in the adult brain. We further characterized the abnormal cell density evolution at $+1/2$ and $-1/2$ topological defects, and proposed a universal mechanism indicating that similar situations can arise when active force coexists with anisotropic friction. Our findings show how topological defects can affect collective cell dynamics, and demonstrate how phenomenological descriptions of active matter can yield understanding of purely physical consequences in multicellular systems.

Online Content Methods, along with any additional Extended Data display items and Source Data, are available in the online version of the paper; references unique to these sections appear only in the online paper.

Received 20 May 2016; accepted 30 March 2017.
Published online 12 April 2017.

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Supplementary Information is available in the online version of the paper.

Acknowledgements We thank H. Chaté, H. Kori, K. Nagai, A. Isomura, K. Takeuchi, D. Nishiguchi, T. Yamamoto, and members of the Kageyama and Sano laboratories for discussions. We acknowledge WPI-iCeMS, Kyoto University, for technical help with flow cytometry. We thank A. M. Klein for supplying equipment and reagents for additional experiments. This work was supported by Core Research for Evolutional Science and Technology (JPMJCR12W2, K.K. and R.K.) and by KAKENHI (numbers 25103004 “Fluctuation & Structure” and 16H06480) from MEXT, Japan (K.K., R.K. and M.S.), and the Platform for Dynamic Approaches to Living Systems from MEXT, Japan (R.K.). K.K. acknowledges two Grants-in-Aid for JSPS Fellows (numbers 24-8031and 28-908).

Author Contributions K.K., R.K. and M.S. designed the project and wrote the manuscript. K.K. performed the experiments and developed the image analysis methods. K.K. and M.S. developed the theory.

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Reviewer Information *Nature* thanks A. Bausch, D. Discher, L. Hirst and the other anonymous reviewer(s) for their contribution to the peer review of this work.

METHODS

No statistical methods were used to predetermine sample size. The experiments were not randomized. The investigators were not blinded to allocation during experiments and outcome assessment.

Cell lines and culture methods. We used cell lines of NPCs, which were previously established from mice of the standard strain E14 ICR[6,33]. We authenticated these cells by confirming that they have a potential to give rise to neurons, oligodendrocytes and astrocytes. We also authenticated C2C12 cells by confirming that they have a potential to differentiate into muscle cells. We did not test for mycoplasma contamination.

Two-dimensional NPC culture was prepared and passaged following the previously described protocols[6,33] with a few modifications. Cells were cultured using DMEM/F12 with HEPES (Invitrogen) supplemented with basic fibroblast growth factor (bFGF) (human, 20 ng ml^{-1}, Wako or Invitrogen), epidermal growth factor (EGF) (mouse, 20 ng ml^{-1}, Invitrogen), and N2 plus supplement (R&D). Dishes were coated with laminin (Wako or Invitrogen). Accutase (ICT) was used for regular passaging. Myoblast cell line C2C12 with H2B-mCherry marker (a kind gift from A. Isomura) was cultured in DMEM (Wako) with 10% serum.

Generation of stable NPC lines. Stable cell lines of NPCs expressing nucleus markers (H2B-mCherry under phosphoglycerate kinase (PGK) promoter) were established using the Nepagene electroporator, by co-electroporating with plasmids carrying Tol2 transposase[34] and anti-Hygromycin (all plasmids were kind gifts from A. Isomura). Electroporated cells were harvested for about 1 week and selected by antibiotics or sorted with BD Aria II.

Imaging. For long time-lapse experiments, NPCs were plated onto laminin-coated glass base dishes (Iwaki) or plastic wells (Falcon) the day before at desired cell densities. Tiled fluorescent and phase contrast imaging was performed by AF6000, AF7000 (Leica Microsystems) or Eclipse Ti-E (Nikon) with Zyla (Andor). C2C12 culture was similarly prepared but without the laminin coating. Experiments with artificial boundaries in the culture (Fig. 3) were conducted by scraping the plastic dish with a razor before applying the culture media.

Immunohistochemistry. Cells were fixed in paraformaldehyde (4% in PBS), and washed three times in PBS before blocking with PBS containing 0.5% BSA and 0.3% Triton for 30 min. The blocking buffer was used for the application of primary antibodies: anti-NCAM (Millipore MAB5324, rabbit, 250×, overnight), anti-PSA-NCAM (DSHB 5A5 concentrate, mouse, 100×, overnight), anti-Nestin (DSHB rat-401, mouse, 100×, 60 min), and secondary antibodies: Alexa Fluor 488 anti-rabbit and anti-mouse (Invitrogen, 500×, 60 min).

Knockdown experiments. For shRNA knockdown experiments (Extended Data Fig. 3d), we used the third-generation lentivirus packaging system. We used pLKO.1 plasmids with anti-NCAM (TRCN0000113653, Sigma) and scrambled sequence. Lentiviruses were harvested for 2 days in Optimem (Invitrogen) without FBS from transfected HEK293T cells (Lipofectamine 3000, Invitrogen). Supernatant with anti-NCAM and scrambled (negative control) viruses were freeze-stored, thawed and N2 plus supplement, bFGF, and EGF were added in the same concentration as the growth media just before applying to near-confluent NPCs. After 24 h incubation, lentivirus-infected NPCs were passaged to a new dish with fresh culture media. The next day, we started antibiotic selection by switching media to fresh growth media containing puromycin (0.5 μg ml^{-1}). Cells were incubated for two days (enough to kill 100% of the non-infected NPCs under the same initial density), passaged to new dishes and harvested for at least two days under growth media without puromycin before imaging or immunostaining.

Summary of image analysis. Colour maps of phase on the cell alignment images and the automatic detection of topological defect points were obtained by a MATLAB script implementing the tensor method[35,36]. Briefly, the gradient tensor of the intensity matrix was calculated for the raw phase contrast images, and the principal angles and nematic orders were calculated from these tensors. See Supplementary Information for further detail.

To count the cells in the fluorescent images for Fig. 2a, Extended Data Figs 1a, b and 5a, the raw images were first median filtered (2 pixels = 1.8 μm). After suppressing small maxima (imhmax in MATLAB with a manually set threshold), the number of regional maxima (imregionalmax in MATLAB) was counted.

For the tracking of cells for Figs 1a, 2b–d, 3b and 4b, d, f, and Extended Data Figs 2a–c, 4a–c, 7c, e, 9a and 10 we developed a nucleus marker tracker (MATLAB script). The position of a cell was defined by the centroid of the cell nucleus fluorescent signal, and was traced in the subsequent time frames based on a cost function of position and the fluorescent intensity. Lost cells or cells tracked but with high value in cost function were excluded from the data analyses. To obtain long and precise single cell tracking data for Fig. 2b–d, Extended Data Figs 2a–c and 4a–c, we further added a manual option in our algorithm to detect failure of tracking and halt to let the operator manually point to the correct position of the cell.

Code and data availability. All MATLAB codes and the data that support the findings of this study are available from the corresponding author (K.K.) upon request.

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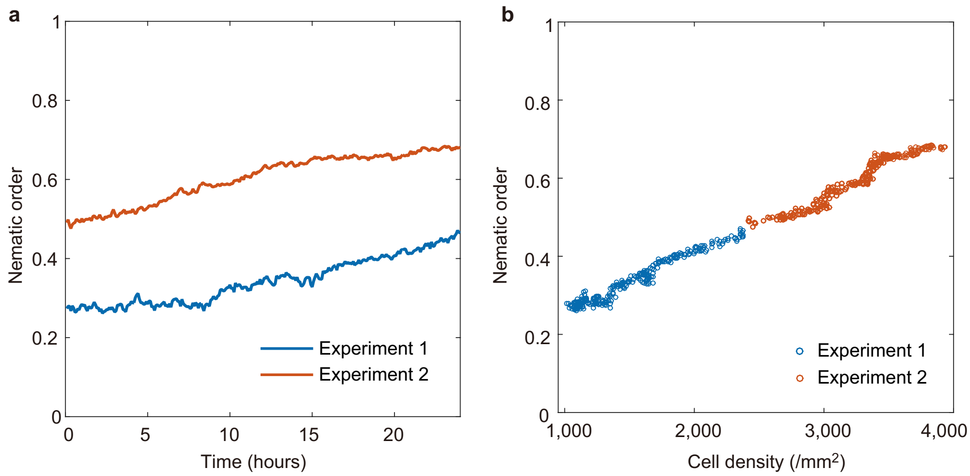
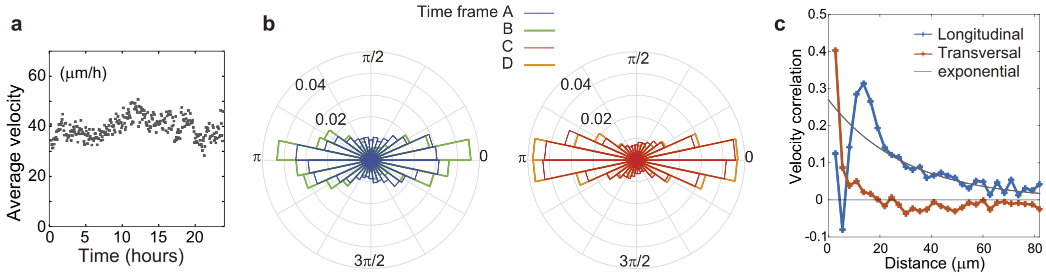
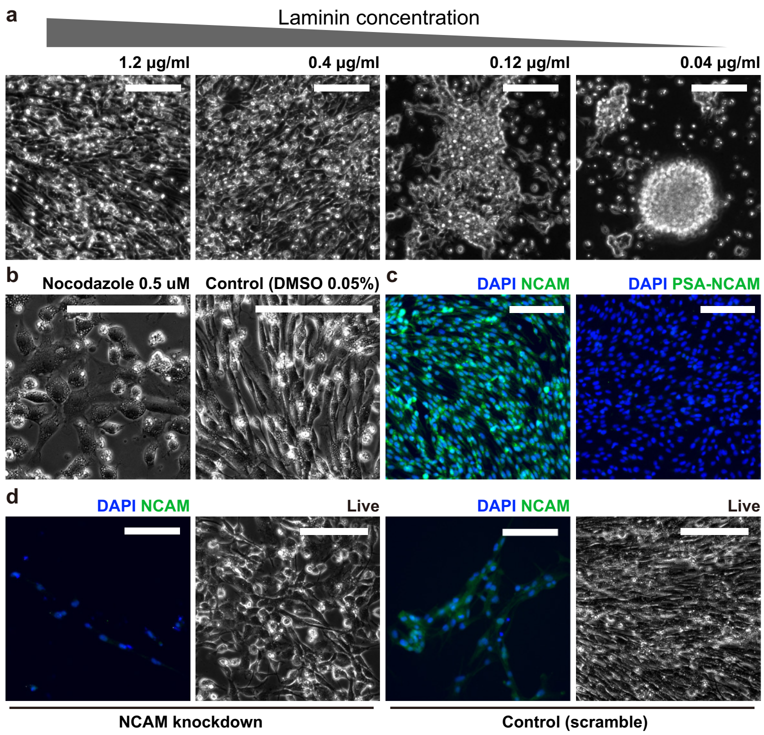


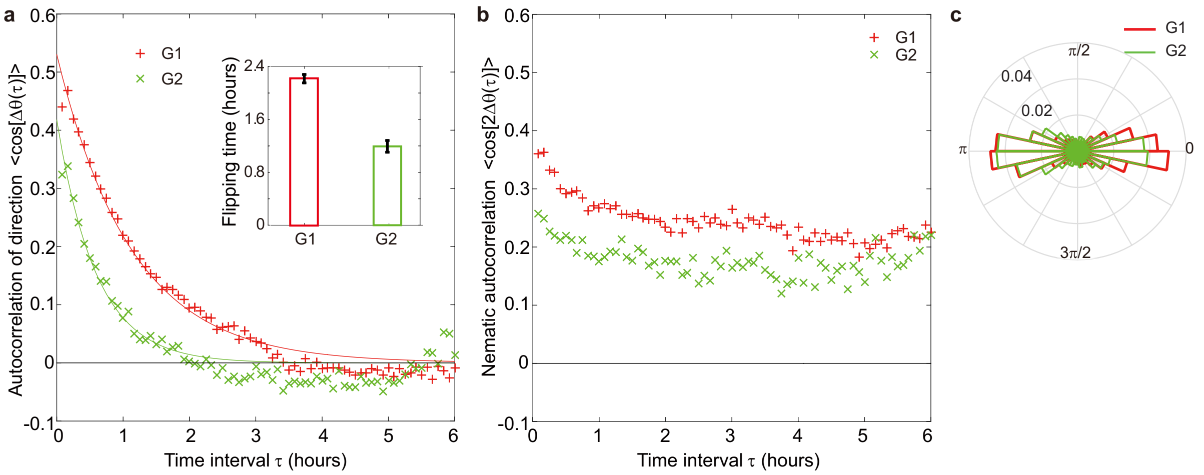
Extended Data Figure 1 | Growing nematic order in NPC culture. a, Nematic order (see equation (S18) in Supplementary Information) calculated for a fixed system size $L = 185\,\mu m$ (chosen as an intermediate size that is larger than a single cell and smaller than the typical distance between defects) for each time frame in the two separate experiments. **b**, Cell density versus nematic order.



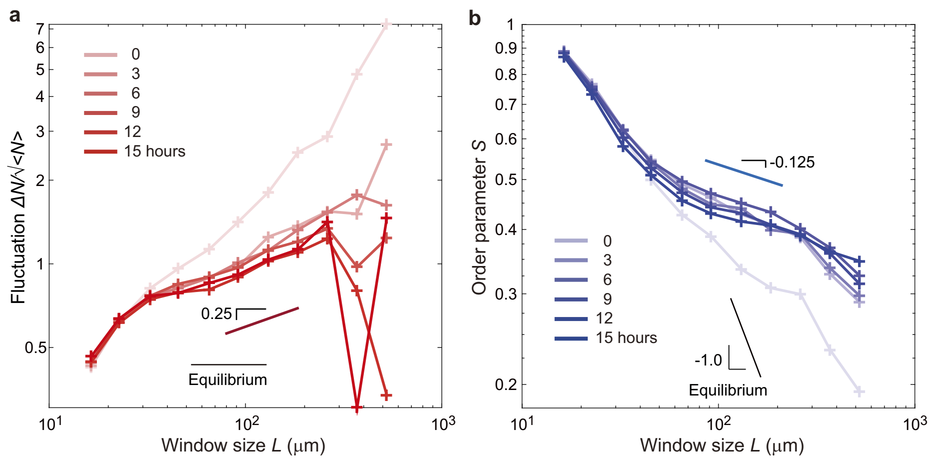
Extended Data Figure 2 | Velocity and angle of cell motion. a, Time-dependence of average cell velocity calculated from the displacement of cells between frames from experiment 2 (high cell density, time frames C and D shown in Fig. 2a). **b**, Distribution of the direction of motion of cells with respect to the longitudinal direction calculated for each cell trajectory. See Supplementary Information for details. **c**, Spatial correlation of velocity in the direction of alignment calculated from the displacements in ordered regions (size about 0.2 mm^2, cropped from images in time frames C and D in Fig. 2a). Longitudinal: correlation of velocity in the direction of alignment between two cells that are apart in the direction of the alignment ($C_v^{\mathrm{L}}(r)$ in Supplementary Information). Transversal: correlation of velocity in the direction of alignment between two cells that are apart in the direction perpendicular to the alignment ($C_v^{\mathrm{T}}(r)$ in Supplementary Information). Exponential: the line is a guide to the eye with values proportional to e^{-r/r_0} with length scale $r_0 = 35\,\mu\mathrm{m}$.



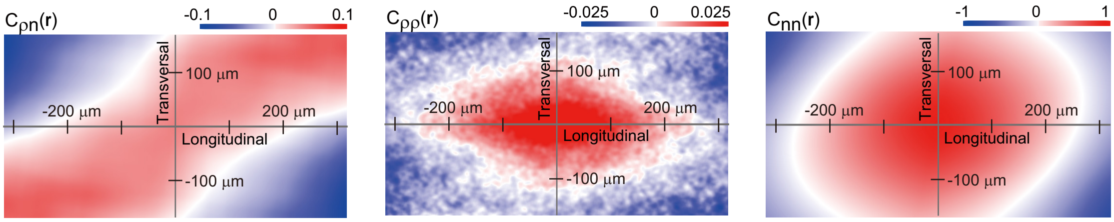
Extended Data Figure 3 | Pattern disruption by low laminin concentration, nocodazole treatment, and NCAM knockdown. **a**, Lowering laminin concentration in the laminin-coated culture changes the behaviour of the NPCs. Cells tend to aggregate when laminin is low. **b**, Nocodazole (a microtubule polymerization inhibitor)-treated NPCs (24 h), showing disrupted pattern, consistent with the bipolar shape of the NPCs being supported by microtubules. **c**, Immunocytochemistry of NPCs. Left, NCAM. Right, PSA-NCAM. NPCs express NCAM but not its polysialylated form. **d**, Pattern of NPCs under stable NCAM knockdown by shRNA lentivirus (the control is shRNA with a non-specific scrambled sequence). See also Supplementary Video 4. Scale bars, 100 µm.



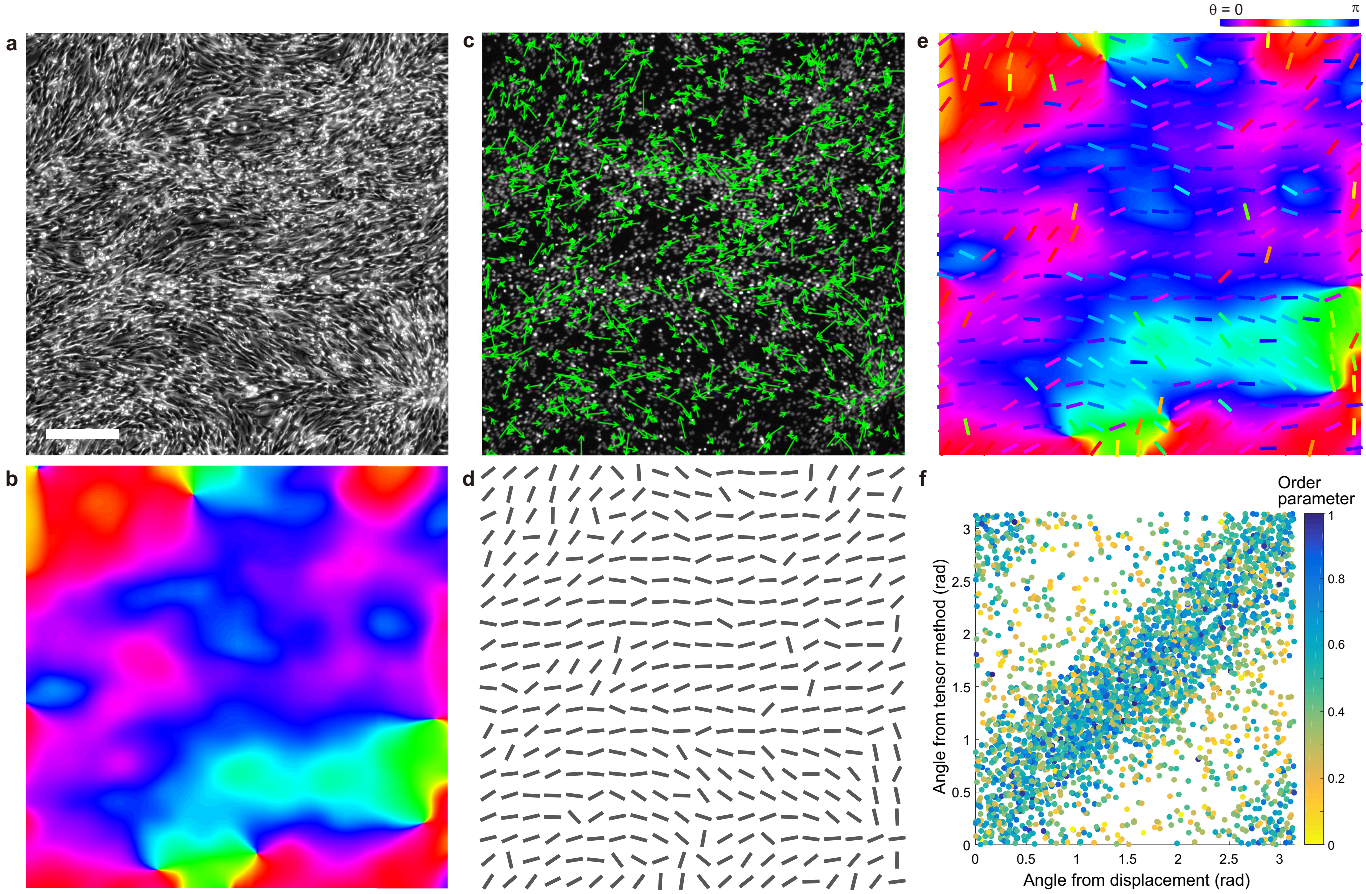
Extended Data Figure 4 | Reversal of the direction of motion during the same cell cycle phase. a–c, Autocorrelations (**a**), nematic autocorrelations (**b**) (where angle brackets indicate the ensemble average), and distribution of velocity angle (**c**) relative to the longitudinal angle calculated for cells in different cell cycle phases (G1 and G2) from the same experiment of Fucci-labelled NPCs at high cell density (>3,000 cells mm^{-2}). The inset in **a** is the flipping time quantified from the fitting of autocorrelation by an exponential function (solid lines in **a**). Error bars correspond to the standard error of fitting. See also Supplementary Video 6.



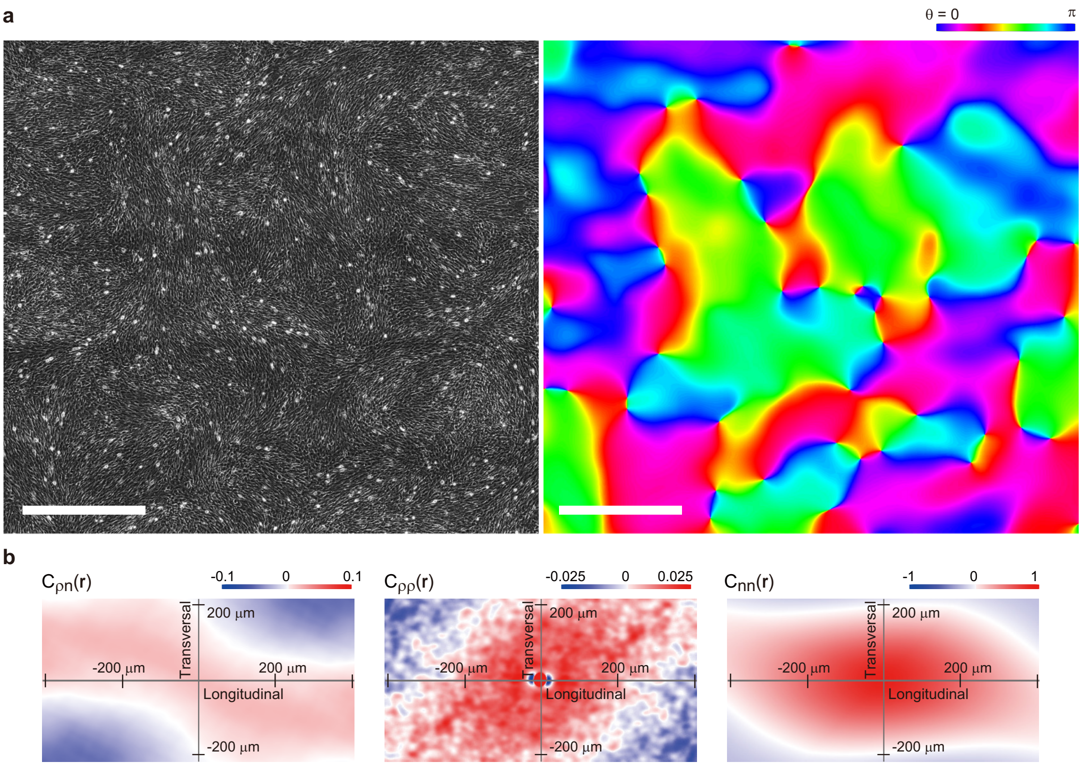
Extended Data Figure 5 | Giant number fluctuation and global order parameter in regions without defect. a, Giant cell number N fluctuation in the dense culture (size about 0.2 mm^2, cropped from images in time frames C and D in Fig. 2a). The fluctuation rescaled by the square root of the mean cell number in the different defined system sizes was calculated from counting of the cell nuclei. Cell density has grown from 3,000 mm^{-2} to 4,500 mm^{-2} during the 15-h period. We find $\Delta N \propto \langle N \rangle^{\alpha}$ (where ΔN is the fluctuation and $\langle N \rangle$ is the average cell number within the fixed size window) with an exponent α larger than 0.7 in the straight region where the system size is larger than the cell and smaller than the maximum window size. **b**, Global nematic order as a function of system size (see equation (S18) in Supplementary Information) in the same time frames as **a**. Nematic order was calculated from the velocity field obtained by the global tracking algorithm. The solid lines are guides to the eye.



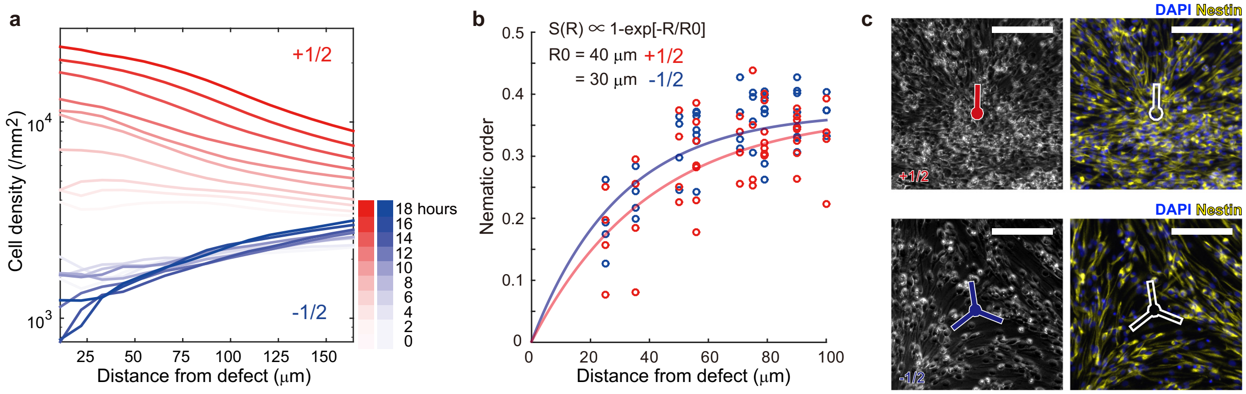
Extended Data Figure 6 | Spatial correlation of the fluctuation of density and alignment. Spatial (two-dimensional displacement ***r***) correlations of density fluctuation and alignment fluctuation ($C_{\rho n}$, $C_{\rho\rho}$ and C_{nn}) calculated in the ordered region without defect (size about 0.2 mm^2, cropped from images in time frames C and D in Fig. 2a). The colours indicate the values of the correlations (red is positive and blue is negative), where the values are normalized using the standard deviations of the density and alignment fluctuations (see equations (S13)–(S15) in Supplementary Information for definitions). The horizontal axis corresponds to the alignment direction defined in the analysed regions. The correlation is the average value calculated from densities and alignments at four separate regions.



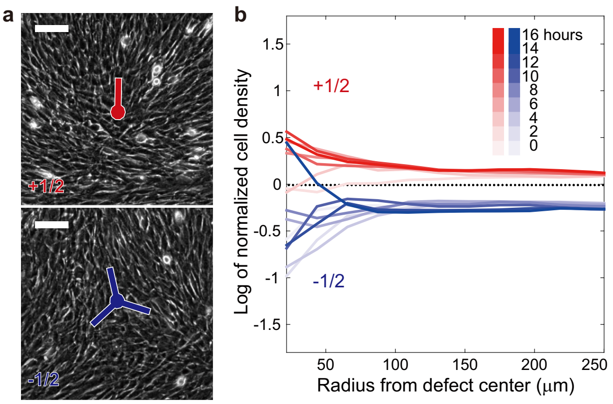
Extended Data Figure 7 | Comparing the tensor method and the tracking method to extract alignment from image. a, Phase contrast image of NPC culture in the high-density regime. Scale bar, 200 μm. **b**, Direction of alignment obtained by applying the tensor method on the phase contrast image data. **c**, Fluorescent image of the cell nucleus (white) in the same position as **a**, overlaid with the displacements (green arrows) in a single time frame (5-min interval). Only about 20% of the calculated displacements (elongated 15-fold) are shown, for the sake of visualization. **d**, Alignment field obtained from the nematic order calculated within the boxes (60 pixels × 60 pixels = 54.7 × 54.7 μm^2) using the displacement vectors. **e**, Merge of the alignment fields obtained by two independent methods. Bars indicating the direction of alignment (same as **d**) were coloured according to the same rule as **b** to see a match between the two fields. **f**, Correlation of the angle of alignment calculated using displacement vectors (**d**) versus the tensor method (**b**). Each dot corresponds to an angle calculated within a box at the same time frame. Eight time frames with interval 2 h from four different positions were used. The colour shows the nematic order parameter in each box.



Extended Data Figure 8 | Patterning and correlations in myoblast cell culture C2C12. a, Large-field image of C2C12 culture. Colour (right) indicates the angle of the local alignment calculated using the tensor method (see Supplementary Information) from the phase contrast image (left). Scale bar, 1 mm. **b**, Spatial correlation of the fluctuation of density and alignment in C2C12 calculated in the ordered region without defect. See equations (S13)–(S15) in Supplementary Information for definitions.



Extended Data Figure 9 | Cell density evolution and nematic order around defects. a, Raw cell density evolution quantified without normalization. The decrease of cell density around the −1/2 defects is unclear compared with the case of normalized cell density (see Fig. 4h), owing to the competition of the outflow and the growth of cells. **b**, Fit of the nematic order near the defect calculated from the data points obtained by the tracking method (corresponding to Fig. 4c). A radially symmetric form of the nematic order is assumed (see Supplementary Information), and $R0$ is the fitted length scale of the defect. Although the nematic order should converge to 1 at large R, the value calculated here is small owing to finite binning (23 μm) and the small fluctuations in the alignment angle over time and independent defects. **c**, Phase contrast (left) and immunofluorescent images (right) of fixed NPCs around topological defects. Nestin (a neural stem cell marker) is universally expressed, meaning that there is no noticeable cell differentiation occurring around the defects. Scale bars, 100 μm.



Extended Data Figure 10 | Absence of cell accumulation and escape around defects in C2C12. **a**, Phase contrast image of C2C12 cells, highlighting the existence of +1/2 and −1/2 topological defects in the ordered pattern. **b**, Cell density dynamics around the defects. Average of four defects each from the same dish over the same time course.